

Student service ddl

```
CREATE DATABASE IF NOT EXISTS studentdb;  
USE studentdb;
```

```
CREATE TABLE students (  
    student_id VARCHAR(10) PRIMARY KEY,  
    first_name VARCHAR(50) NOT NULL,  
    last_name VARCHAR(50) NOT NULL,  
    email VARCHAR(100) NOT NULL UNIQUE,  
    tel_number VARCHAR(20) NOT NULL,  
    programme_code VARCHAR(20) NOT NULL,  
    status ENUM('ACTIVE','INACTIVE') NOT NULL DEFAULT 'ACTIVE'  
);
```

```
CREATE TABLE academic_records (  
    record_id INT AUTO_INCREMENT PRIMARY KEY,  
    student_id VARCHAR(10) NOT NULL,  
    academic_year VARCHAR(9) NOT NULL,  
    semester INT NOT NULL,  
    gpa DECIMAL(3,2) NOT NULL,  
    cgpa DECIMAL(3,2) NOT NULL,  
    credits_earned INT NOT NULL DEFAULT 0,
```

```
    CONSTRAINT fk_student  
        FOREIGN KEY (student_id)  
        REFERENCES students(student_id),
```

```
    CONSTRAINT chk_semester  
        CHECK (semester BETWEEN 1 AND 3),
```

```
    CONSTRAINT chk_gpa  
        CHECK (gpa BETWEEN 0.00 AND 4.00),
```

```
    CONSTRAINT chk_cgpa  
        CHECK (cgpa BETWEEN 0.00 AND 4.00),
```

```
    CONSTRAINT chk_credits  
        CHECK (credits_earned >= 0)  
);
```

Student service dml

INSERT INTO students VALUES

```
('S1001','John','Tan','john.tan@student.edu','0123456781','CS110','ACTIVE'),
('S1002','Sarah','Lee','sarah.lee@student.edu','0123456782','CS110','ACTIVE'),
('S1003','Kevin','Lim','kevin.lim@student.edu','0123456783','SE220','ACTIVE'),
('S1004','Amanda','Wong','amanda.wong@student.edu','0123456784','SE220','ACTIVE'),
('S1005','Jason','Ng','jason.ng@student.edu','0123456785','IT330','ACTIVE'),
('S1006','Melissa','Chong','melissa.chong@student.edu','0123456786','IT330','ACTIVE'),
('S1007','Daniel','Ong','daniel.ong@student.edu','0123456787','CS110','ACTIVE'),
('S1008','Rachel','Yap','rachel.yap@student.edu','0123456788','SE220','ACTIVE'),
('S1009','Bryan','Khoo','bryan.khoo@student.edu','0123456789','IT330','ACTIVE'),
('S1010','Grace','Teh','grace.teh@student.edu','0123456790','CS110','ACTIVE');
```

INSERT INTO academic_records

(student_id, academic_year, semester, gpa, cgpa, credits_earned)
VALUES

```
('S1001','2024/2025',1,3.45,3.45,18),
('S1001','2024/2025',2,3.60,3.53,36),
```

```
('S1002','2024/2025',1,3.20,3.20,18),
('S1002','2024/2025',2,3.40,3.30,36),
```

```
('S1003','2024/2025',1,3.75,3.75,18),
('S1003','2024/2025',2,3.82,3.79,36),
```

```
('S1004','2024/2025',1,3.55,3.55,18),
('S1004','2024/2025',2,3.61,3.58,36),
```

```
('S1005','2024/2025',1,3.10,3.10,18),
('S1005','2024/2025',2,3.25,3.18,36),
```

```
('S1006','2024/2025',1,3.90,3.90,18),
('S1006','2024/2025',2,3.88,3.89,36),
```

```
('S1007','2024/2025',1,3.33,3.33,18),
('S1007','2024/2025',2,3.47,3.40,36),
```

```
('S1008','2024/2025',1,3.68,3.68,18),
('S1008','2024/2025',2,3.72,3.70,36),
```

```
('S1009','2024/2025',1,3.25,3.25,18),
('S1009','2024/2025',2,3.35,3.30,36),
```

```
('S1010','2024/2025',1,3.80,3.80,18),
('S1010','2024/2025',2,3.85,3.83,36);
```

Course enrollment service ddl

```
CREATE DATABASE IF NOT EXISTS enrolmentdb;  
USE enrolmentdb;
```

```
CREATE TABLE courses (  
    course_id VARCHAR(10) PRIMARY KEY,  
    course_name VARCHAR(100) NOT NULL,  
    credit_hours INT NOT NULL,  
    max_capacity INT NOT NULL,  
    available_seats INT NOT NULL,
```

```
    CONSTRAINT chk_capacity  
        CHECK (max_capacity > 0),
```

```
    CONSTRAINT chk_available  
        CHECK (available_seats >= 0)
```

```
);
```

```
CREATE TABLE enrollments (  
    enrollment_id INT AUTO_INCREMENT PRIMARY KEY,  
  
    student_id VARCHAR(10) NOT NULL,  
  
    course_id VARCHAR(10) NOT NULL,  
  
    enrollment_date DATETIME DEFAULT CURRENT_TIMESTAMP,  
  
    status ENUM(  
        'ENROLLED',  
        'DROPPED',  
        'COMPLETED'  
    ) DEFAULT 'ENROLLED',  
  
    FOREIGN KEY (course_id)  
        REFERENCES courses(course_id)  
);
```

Course service dml

```
INSERT INTO courses VALUES  
('CSC101','Programming Fundamentals',3,50,40),  
('CSC102','Database Systems',3,50,42),  
('CSC103','Computer Networks',3,45,30),  
('CSC104','Operating Systems',3,40,25),  
('CSC105','Software Engineering',3,50,35),  
('CSC106','Web Development',3,60,50),  
('CSC107','Data Structures',3,55,44),
```

```
('CSC108','Cyber Security',3,40,28),  
( 'CSC109','Artificial Intelligence',3,35,20),  
( 'CSC110','Cloud Computing',3,45,30);
```

```
INSERT INTO enrollments  
(student_id, course_id, status)  
VALUES  
( 'S1001','CSC101','ENROLLED'),  
( 'S1001','CSC102','ENROLLED'),  
  
( 'S1002','CSC101','ENROLLED'),  
( 'S1002','CSC103','ENROLLED'),  
  
( 'S1003','CSC104','ENROLLED'),  
( 'S1003','CSC105','ENROLLED'),  
  
( 'S1004','CSC106','ENROLLED'),  
( 'S1004','CSC107','ENROLLED'),  
  
( 'S1005','CSC108','ENROLLED'),  
( 'S1005','CSC109','ENROLLED'),  
  
( 'S1006','CSC110','ENROLLED'),  
( 'S1006','CSC101','ENROLLED'),  
  
( 'S1007','CSC102','ENROLLED'),  
( 'S1007','CSC103','ENROLLED'),  
  
( 'S1008','CSC104','ENROLLED'),  
( 'S1008','CSC105','ENROLLED'),  
  
( 'S1009','CSC106','ENROLLED'),  
( 'S1009','CSC107','ENROLLED'),  
  
( 'S1010','CSC108','ENROLLED'),  
( 'S1010','CSC109','ENROLLED');
```

Notification ddl

```
CREATE DATABASE IF NOT EXISTS notificationdb;  
USE notificationdb;
```

```
CREATE TABLE notifications (  
    notification_id INT AUTO_INCREMENT PRIMARY KEY,  
  
    student_id VARCHAR(10) NOT NULL,  
  
    enrolment_id INT,  
  
    message VARCHAR(255) NOT NULL,  
  
    notification_type ENUM(  
        'EMAIL',  
        'SMS',  
        'PUSH'  
    ) NOT NULL,  
  
    sent_time DATETIME DEFAULT CURRENT_TIMESTAMP,  
  
    status ENUM(  
        'PENDING',  
        'SENT',  
        'FAILED'  
    ) DEFAULT 'SENT'  
);
```

Notification dml

```
INSERT INTO notifications  
(student_id, enrolment_id, message, notification_type, status)  
VALUES  
(  
'S1001',1,'Enrolment confirmed for CSC101','EMAIL','SENT'),  
(  
'S1001',2,'Enrolment confirmed for CSC102','EMAIL','SENT'),  
(  
'S1002',3,'Enrolment confirmed for CSC101','SMS','SENT'),  
(  
'S1002',4,'Enrolment confirmed for CSC103','EMAIL','SENT'),  
(  
'S1003',5,'Enrolment confirmed for CSC104','PUSH','SENT'),  
(  
'S1003',6,'Enrolment confirmed for CSC105','EMAIL','SENT'),  
(  
'S1004',7,'Enrolment confirmed for CSC106','EMAIL','SENT'),  
(  
'S1004',8,'Enrolment confirmed for CSC107','SMS','SENT'),  
(  
'S1005',9,'Enrolment confirmed for CSC108','EMAIL','SENT'),  
(  
'S1005',10,'Enrolment confirmed for CSC109','PUSH','SENT'),  
(  
'S1006',11,'Enrolment confirmed for CSC110','EMAIL','SENT'),  
(  
'S1006',12,'Enrolment confirmed for CSC101','EMAIL','SENT'),  
(  
'S1007',13,'Enrolment confirmed for CSC102','SMS','SENT'),  
(  
'S1007',14,'Enrolment confirmed for CSC103','EMAIL','SENT'),  
(  
'S1008',15,'Enrolment confirmed for CSC104','EMAIL','SENT'),  
(  
'S1008',16,'Enrolment confirmed for CSC105','PUSH','SENT'),
```

```
('S1009',17,'Enrolment confirmed for CSC106','EMAIL','SENT'),  
('S1009',18,'Enrolment confirmed for CSC107','SMS','SENT'),  
('S1010',19,'Enrolment confirmed for CSC108','EMAIL','SENT'),  
('S1010',20,'Enrolment confirmed for CSC109','EMAIL','SENT');
```

Two changes made in database

Notification Service

```
ALTER TABLE notifications ADD COLUMN reservation_id INT NULL;  
ALTER TABLE notifications ADD COLUMN room_reservation_id INT NULL;
```

Course EnrollmentService

```
ALTER TABLE enrollments ADD COLUMN created_at DATETIME DEFAULT  
CURRENT_TIMESTAMP;
```

Library and booking ddl

```
CREATE DATABASE IF NOT EXISTS librarydb;  
USE librarydb;
```

```
CREATE TABLE books (  
    book_id VARCHAR(10) PRIMARY KEY,  
    title VARCHAR(150) NOT NULL,  
    author VARCHAR(100) NOT NULL,  
    isbn VARCHAR(20) UNIQUE,  
    availability_status ENUM(  
        'AVAILABLE',  
        'BORROWED'  
    ) DEFAULT 'AVAILABLE'  
);
```

```
CREATE TABLE book_reservations (  
    reservation_id INT AUTO_INCREMENT PRIMARY KEY,  
  
    book_id VARCHAR(10) NOT NULL,  
  
    student_id VARCHAR(10) NOT NULL,  
  
    reservation_date DATETIME DEFAULT CURRENT_TIMESTAMP,  
  
    status ENUM(  

```

```

        'ACTIVE',
        'COMPLETED',
        'CANCELLED'
    ) DEFAULT 'ACTIVE',

    FOREIGN KEY (book_id)
        REFERENCES books(book_id)
);

CREATE TABLE rooms (
    room_id VARCHAR(10) PRIMARY KEY,
    room_name VARCHAR(100) NOT NULL,
    capacity INT NOT NULL,
    availability_status ENUM(
        'AVAILABLE',
        'BOOKED'
    ) DEFAULT 'AVAILABLE'
);

CREATE TABLE room_reservations (
    room_reservation_id INT AUTO_INCREMENT PRIMARY KEY,

    room_id VARCHAR(10) NOT NULL,

    student_id VARCHAR(10) NOT NULL,

    reservation_date DATE NOT NULL,

    start_time TIME NOT NULL,

    end_time TIME NOT NULL,

    status ENUM(
        'ACTIVE',
        'COMPLETED',
        'CANCELLED'
    ) DEFAULT 'ACTIVE',

    FOREIGN KEY (room_id)
        REFERENCES rooms(room_id)
);

book dml
INSERT INTO books VALUES
('B001','Database Systems','Thomas Connolly','978001','AVAILABLE'),
('B002','Computer Networks','Andrew Tanenbaum','978002','AVAILABLE'),
('B003','Operating Systems','Abraham Silberschatz','978003','AVAILABLE'),
('B004','Software Engineering','Ian Sommerville','978004','AVAILABLE'),

```

```
('B005','Artificial Intelligence','Stuart Russell','978005','AVAILABLE'),
('B006','Cloud Computing','Rajkumar Buyya','978006','AVAILABLE'),
('B007','Java Programming','Herbert Schildt','978007','AVAILABLE'),
('B008','Cyber Security Basics','William Stallings','978008','AVAILABLE'),
('B009','Web Development','Jon Duckett','978009','AVAILABLE'),
('B010','Data Structures','Mark Allen Weiss','978010','AVAILABLE');
```

Book reservation dml

```
INSERT INTO book_reservations
```

```
(book_id, student_id, status)
```

```
VALUES
```

```
('B001','S1001','ACTIVE'),
('B002','S1002','ACTIVE'),
('B003','S1003','ACTIVE'),
('B004','S1004','ACTIVE'),
('B005','S1005','ACTIVE'),
('B006','S1006','ACTIVE'),
('B007','S1007','ACTIVE'),
('B008','S1008','ACTIVE'),
('B009','S1009','ACTIVE'),
('B010','S1010','ACTIVE');
```

Room dml

```
INSERT INTO rooms VALUES
```

```
('R001','Discussion Room A',6,'AVAILABLE'),
('R002','Discussion Room B',8,'AVAILABLE'),
('R003','Discussion Room C',10,'AVAILABLE'),
('R004','Discussion Room D',6,'AVAILABLE'),
('R005','Discussion Room E',12,'AVAILABLE');
```

Room Reservation dml

```
INSERT INTO room_reservations
```

```
(room_id, student_id, reservation_date, start_time, end_time, status)
```

```
VALUES
```

```
('R001','S1001','2025-06-01','09:00:00','11:00:00','ACTIVE'),
('R002','S1002','2025-06-01','10:00:00','12:00:00','ACTIVE'),
('R003','S1003','2025-06-02','08:00:00','10:00:00','ACTIVE'),
('R004','S1004','2025-06-02','13:00:00','15:00:00','ACTIVE'),
('R005','S1005','2025-06-03','09:00:00','11:00:00','ACTIVE'),
('R001','S1006','2025-06-03','14:00:00','16:00:00','ACTIVE'),
('R002','S1007','2025-06-04','09:00:00','11:00:00','ACTIVE'),
('R003','S1008','2025-06-04','15:00:00','17:00:00','ACTIVE'),
('R004','S1009','2025-06-05','10:00:00','12:00:00','ACTIVE'),
('R005','S1010','2025-06-05','13:00:00','15:00:00','ACTIVE');
```


Reporting and analytics ddl

```
CREATE DATABASE IF NOT EXISTS analyticsdb;  
USE analyticsdb;
```

```
CREATE TABLE report (  
    report_id INT AUTO_INCREMENT PRIMARY KEY,  
  
    course_id VARCHAR(10) NOT NULL,  
  
    total_enrolments INT NOT NULL,  
  
    semester VARCHAR(20) NOT NULL,  
  
    generated_date DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

```
CREATE TABLE analytics (  
    analytics_id INT AUTO_INCREMENT PRIMARY KEY,  
  
    service_name VARCHAR(100) NOT NULL,  
  
    total_requests INT NOT NULL,  
  
    successful_requests INT NOT NULL,  
  
    generated_date DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

Report dml

```
INSERT INTO report  
(course_id,total_enrolments,semester)  
VALUES  
('CSC101',15,'2024/2025-S2'),  
('CSC102',12,'2024/2025-S2'),  
('CSC103',10,'2024/2025-S2'),  
('CSC104',8,'2024/2025-S2'),  
('CSC105',14,'2024/2025-S2'),  
('CSC106',18,'2024/2025-S2'),  
('CSC107',13,'2024/2025-S2'),  
('CSC108',11,'2024/2025-S2'),  
('CSC109',9,'2024/2025-S2'),  
('CSC110',7,'2024/2025-S2');
```

Analytics dml

```
INSERT INTO analytics  
(service_name,total_requests,successful_requests)  
VALUES  
('Student Profile Service',5000,4950),
```

('Course Enrolment Service',4800,4750),
('Notification Service',4500,4480),
('Library Service',3000,2985),
('Analytics Service',2500,2490),
('Student Profile Service',5200,5175),
('Course Enrolment Service',4900,4860),
('Notification Service',4600,4585),
('Library Service',3100,3090),
('Analytics Service',2600,2592);